

DAiR Radar – Device and Operation Summary

Reichman UAV Radar Project

March 22, 2026

DAiR Radar – Device and Operation Summary

What Is Radar?

- Definition: Radar (Radio Detection And Ranging) transmits radio waves and measures their echoes from objects to determine distance (range), direction (angle), and motion (radial velocity).
- Core idea (end-to-end):
 - Transmit (Tx) a known waveform (e.g., short pulses or a frequency-modulated “chirp”).
 - Energy reflects off objects (“targets”) and returns as echoes.
 - Receive (Rx) the echoes, amplify/filter, down-convert, and digitize them.
 - Digital processing estimates range, angle, and Doppler (radial velocity), then detects targets above noise/clutter.
- Roles of Tx and Rx:
 - Tx: Generates power and shapes where energy goes (beam/coverage). The signal bandwidth largely sets range resolution; average power and duty cycle affect maximum detection range.
 - Rx: Collects echoes. With an array and digital beamforming, multiple simultaneous receive beams can “listen” in many directions. Receiver noise figure and gain impact sensitivity (SNR).
- Range measurement (FMCW and pulse-compression basics):
 - Pulsed/pulse-compression radars transmit short, wide-band pulses (often chirped) and use a matched filter to compress echoes in time. Range comes from the echo delay τ with $R \approx c\tau/2$ (where c is the speed of light). Range resolution $\Delta R \approx c/(2B)$, where B is signal bandwidth.
 - FMCW (frequency-modulated continuous wave) radars transmit a linear “chirp” whose instantaneous frequency sweeps at slope S (Hz/s). Mixing the received echo with the transmit replica yields a beat frequency f_b proportional to range: $f_b \approx \frac{2SR}{c}$ (for an up-chirp, zero Doppler). Thus $R \approx \frac{cf_b}{2S}$. In practice, a range FFT over fast-time (samples within a chirp) resolves discrete range bins; data report bin size as `RgBinSz`.
 - Practical notes: Wider bandwidth improves ΔR , longer coherent integration improves SNR; calibration accounts for internal delays so that zero-range maps near $f_b = 0$.
- Motion (Doppler) measurement and f_d :
 - Doppler is the frequency shift caused by target motion toward/away from the radar. For monostatic radar, the Doppler frequency is $f_d \approx \frac{2v}{\lambda}$, where v is radial velocity and $\lambda = c/f_c$ is wavelength at carrier frequency f_c .
 - FMCW separation of range and Doppler: Over many chirps (slow-time), a Doppler FFT recovers f_d while the fast-time FFT yields range. With pure up-chirps, f_b contains both range and Doppler terms; using alternating up/down chirps or processing across chirps separates them. Velocity resolution improves with longer coherent processing (more chirps) and is summarized in data by the Doppler bin size `DopBinSz`.
 - Unambiguous velocity depends on the chirp/PRI timing; approximate bounds scale with wavelength and chirp repetition frequency (higher PRF implies larger unambiguous velocity).
- Angle (azimuth/elevation): From antenna patterns and/or array phase differences via beamforming. Narrower beams yield finer angular resolution; element-level digitization enables forming multiple Rx beams simultaneously.

-
- From echoes to tracks:
 - Detection: Processing (e.g., CFAR) declares “plots” (candidate targets) in range-Doppler-angle cells when energy exceeds local noise/clutter.
 - Tracking: A tracker associates plots over time into “tracks” with IDs and kinematics (position, range-rate), enabling classification and prediction.
 - Detecting movement nuances:
 - Macro Doppler indicates approach/retreat speed; micro-Doppler captures fine modulations (e.g., rotor blades, wing flaps) useful for discriminating UAVs vs. birds.

What The Radar Is

- Name/Type: DAiR – tactical, multi-mission radar.
- Band/Frequency: X-Band, 9-10 GHz (configurable in 5 MHz steps).
- Architecture: Software-defined radar (waveforms and missions configurable via software/firmware parameters).
- Platform support: Accepts INS/GPS; usable on fixed or mobile platforms; fielded as single static panel, four-panel 360° configuration, or as a rotating panel.

Antenna/Array Configuration

- Transmit (TX): Single transmit column with 16 elements forming a wide TX beam covering about 110° azimuth and 6.3° elevation.
- Elevation scanning: $\pm 40^\circ$ by per-element phase control (electronic steering in elevation).
- Receive (RX): Digital at the element level – 20×16 elements (20 columns by 16 elements each) for a total of 320 RX elements, each digitized (A/D), enabling multiple simultaneous RX beams (digital beamforming) to fill the TX beam coverage.

Measured Quantities (from radar returns)

- Range and angle (azimuth/elevation) of targets.
- Radial velocity (Doppler) and Doppler-related features (e.g., Doppler spread).
- RCS (Radar Cross Section) estimates.

Operation Concept

- Track-While-Search (TWS): Periodic search beams enable continuous detection with updates suitable for high-quality tracking.
- One-dimensional elevation scan: The wide TX beam provides large azimuth coverage; volume search is achieved by scanning only in elevation. This allows longer coherent integration for improved Doppler resolution while maintaining high update rates without sacrificing search volume.
- GPU-based processing pipeline: All post-firmware signal processing (including the tracker) runs on GPU, supporting concurrent tracking of thousands of targets without explicit target prioritization.
- Software-defined waveforms: The signal processing chain is parameterized by waveform choice; different missions can be configured without software rewrites.

Performance/Hardware Parameters (as listed)

- TX peak power: 30 W per element (noted effective limit 22.55 W due to cooling).
- Duty cycle (DC): 10%.
- Frequency tuning: 9-10 GHz, 5 MHz steps.
- Gains:
 - Gtx (TX gain): 20 dB.
 - Grx (RX gain): 30 dB (28.8 dB with tapering).
- Beamwidths (no tapering):
 - TX: Elevation 5.7° , Azimuth 58° .

- RX: Elevation 5.7°, Azimuth 4.5°.
- Receiver noise figure (NF): 4 dB.

Outputs and Data Products (relevant to classification)

- Detections (plots) with range/angle/SNR.
- Tracks with unique IDs, kinematics (position, range-rate), and quality/flags.
- Time-Doppler signatures derivable per track (used for discrimination between UAVs and birds).

ICD Data Model (how data is structured)

- Message types
 - CPI_Plots_Report: Per-CPI detections (plots), Range-Doppler (RD) metadata, noise/clutter matrices.
 - CPI_Track_Report: Per-CPI tracks with kinematics and state/flags.
- Common header/timing
 - CPI_Time (μs), AntennaID, CpiType, CpiID, Nppis (PPIs/looks within a CPI), CPILength.
 - CoordinateSystemFlg: 1 = geographic (LLA/ECEF), 0 = platform/antenna frame.
- Resolution and calibration context
 - RgBinSz (meters per range bin), DopBinSz (m/s per Doppler bin).
 - GlobalNoisePerBeam, ClutterAveragePerBeam (quality indicators per beam).
 - DopplerCorrectionFactorsPerBeamAndPPI (apply when comparing Doppler across beams/scans).
- INS/platform/rotation
 - AntPosLLA (lat, lon, alt), Ins_yaw/pitch/roll.
 - BaseLineVecOffsetToAnt (INS-to-antenna offset), CurrRotationModeFlg, CurrRotatingYawSpeed.
- Detections (plots)
 - DetInd, AssociatedTrackIdx.
 - Range, Azimuth, Elevation, SNR_L, SNR_G.
- Tracks
 - IDs/times: TrkID, TrkGlobalIdx, TrkStartTime, LastUpdateTime.
 - Kinematics: X/Y/Z (or XX/YY/ZZ), Vx/Vy/Vz, Ax/Ay/Az.
 - Polar: Range, RangeRate, Azimuth, Elevation.
 - Flags/quality: TrkCategory, Coverage_flag, MnvCount, TrackState bits (e.g., Hovering, MicroDoppler).
- Range-Doppler usage
 - RD cubes/maps are indexed by time/beam/range/doppler; use RgBinSz with the track's Range to select the correct range bin, then read Doppler vs. time for micro-Doppler signatures.
 - Cross-check RangeRate against Doppler peaks (monostatic relation $v \approx (\lambda/2)f_d$; for 9-10 GHz, $\lambda \approx 3.0\text{-}3.3$ cm).

Practical Notes

- Large azimuth coverage with elevation-only scan is a central design choice; expect high update rates and good Doppler resolution for moving targets.
- Multiple simultaneous RX beams allow better coverage and SNR management within the wide TX footprint.
- For algorithm work (classification/tracking): use range, angles, radial velocity, Doppler spread, and RCS where available; the system runs in real time on GPU and can support many concurrent tracked objects.

Glossary (minimal)

- X-Band: Radar operating between ~8-12 GHz; here 9-10 GHz.
- Beamwidth: Angular width of the main lobe (smaller is narrower/more directional).
- Doppler (radial velocity): Frequency shift due to motion toward/away from radar.

-
- RCS: Effective reflecting area; larger RCS usually implies stronger returns.
 - TWS (Track-While-Search): Interleaves search and track updates to maintain tracks while continuing to search.
 - Digital beamforming: Forming multiple beams from element-level digitized channels in receive.